MXB103 Formula Sheet

Everything you need to **memorise**, week by week. Each formula has a short **meaning** and a quick worked example (**e.g.**) underneath. Each week ends with **Remember** notes: the non-formula facts the exam keeps testing. Most examples come from the 2023–2025 exams or the lecture problems, so they double as revision of real questions. The full explanations and examples are in each `weekXX_*.md` file.

Week 1: Errors & floating point (4 formulas)

1.1 Absolute and relative error

$$\text{abs} = |x - \tilde{x}| \qquad \text{rel} = \frac{|x - \tilde{x}|}{|x|}$$

Meaning: x = true value, $\tilde{x}$ = approximation. Relative error $\approx$ the number of correct digits, and it's the one to trust when numbers are very big or very small.

e.g. True value $x = 4.0$, approximation $\tilde{x} = 4.1$:

- abs $= |4.0 - 4.1| = 0.1$
- rel $= \dfrac{0.1}{4.0} = 0.025$

Now shrink both: $x = 4.0 \times 10^{-3}$, $\tilde{x} = 4.1 \times 10^{-3}$:

- abs $= 10^{-4}$ (looks tiny!)
- rel $= \dfrac{10^{-4}}{4.0 \times 10^{-3}} = 0.025$ (exactly as good as before)

▲ **Edge case: the true value is 0.** Relative error is undefined:

- $x = 0$, $\tilde{x} = 10^{-5}$: rel $= \dfrac{10^{-5}}{0}$ is undefined, so report abs $= 10^{-5}$ instead
- always divide by the **true** value $|x|$, never by $|\tilde{x}|$ (a recurring MATLAB bug)

1.2 Smallest positive number in $F(\beta, k, m, M)$

$$S = 1.00\cdots0 \times \beta^m = \beta^m$$

Meaning: Smallest significand (1.000…) times the smallest exponent. Anything smaller underflows to 0.

e.g. $F(2, 5, -4, 4)$, so $\beta = 2$ and $m = -4$ (2025 Q1a):

- smallest significand: $1.0000_2 = 1$
- smallest exponent: 2^{-4}
- $S = 1 \times 2^{-4} = 0.0625$

▲ **Edge case: underflow.** Same system (allowed exponents $-4 \le e \le 4$):

- $x = 0.1 = (1.10011\ldots)_2 \times 2^{-4}$: $e = -4$ is allowed, so it is stored (rounded) ✓
- $x = 0.05 = (1.10011\ldots)_2 \times 2^{-5}$: $e = -5 < m = -4$, so **underflow** and $fl(x) = 0$ ✗
- quick test: $0.05 < S = 0.0625$, so it was always going to underflow

1.3 Largest number

$$L = (\beta - 1).(\beta - 1)\cdots(\beta - 1) \times \beta^M = (\beta - \beta^{1-k})\,\beta^M$$

Meaning: Every digit at its maximum, times the largest exponent. Anything bigger overflows to Inf.

e.g. $F(2, 4, -2, 3)$, so $\beta = 2$, $k = 4$, $M = 3$:

- largest significand: $1.111_2 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = 1.875$
- largest exponent: $2^3 = 8$
- $L = 1.875 \times 8 = 15$
- formula check: $(2 - 2^{1-4}) \times 2^3 = (2 - 0.125) \times 8 = 15$ ✓

▲ **Edge case: overflow.** Same system (allowed exponents $-2 \le e \le 3$):

- $x = 13 = 1101_2 = 1.101_2 \times 2^3$: $e = 3$ is allowed, so it is stored exactly ✓
- $x = 20 = 10100_2 = 1.0100_2 \times 2^4$: $e = 4 > M = 3$, so **overflow** and $fl(x) = \text{Inf}$ ✗
- sneaky case, $x = 15.75 = 1.11111_2 \times 2^3$: $e = 3$ looks fine, but rounding to 4 digits gives $1.111|11 \to 10.000_2$, so x becomes $1.000_2 \times 2^4$. Now $e = 4 > 3$, so **overflow** ✗ (it rounded past $L = 15$)

1.4 Unit roundoff

$$u = \tfrac{1}{2}\,\beta^{1-k} \qquad \text{so} \qquad \frac{|x - fl(x)|}{|x|} \le u$$

Meaning: The worst relative error from rounding a number into F. It depends only on the base and the number of digits.

e.g. $F(3, 6, -7, 8)$, so $\beta = 3$ and $k = 6$ (2024 Q1b):

- $u = \frac{1}{2} \cdot 3^{1-6} = \frac{1}{2} \cdot 3^{-5} = \frac{1}{486} = 0.0021$
- meaning: rounding any number into this system gives a relative error of at most 0.21%

▲ **Edge case: the bound fails when you underflow.** In $F(10, 2, -1, 1)$, $u = \frac{1}{2} \cdot 10^{-1} = 0.05$:

- $fl(0.04) = 0$ (underflow), so the relative error is $\frac{|0.04 - 0|}{0.04} = 1$, far bigger than u
- u only bounds the rounding of numbers **inside** the range $[S, L]$
- a number already in F has zero error, e.g. $fl(0.23) = 0.23$

Remember

- $F(\beta, k, m, M)$ stores $\pm d_1.d_2\cdots d_k \times \beta^e$ with $d_1 \neq 0$ and $m \le e \le M$. Here k counts **all** digits, including d_1.
- Decimal → binary: divide the integer part by 2 (read remainders **upwards**); multiply the fraction by 2 (read integer parts **downwards**).
- $fl(x)$: convert to base β → normalise → check exponent → round to k digits.
- IEEE double = $F(2, 53, -1022, 1023)$, with $u \approx 1.11 \times 10^{-16}$.

Week 2: Cancellation & Taylor (3 formulas + 1 recipe)

2.1 Conjugate trick (removes cancellation)

$$A - B = \frac{A^2 - B^2}{A + B} \qquad \text{e.g.} \qquad \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \frac{-2c}{b + \sqrt{b^2 - 4ac}}$$

Meaning: Subtracting nearly-equal rounded numbers destroys digits. Rewrite the formula so it **adds** instead. Other fixes: $\log A - \log B = \log \frac{A}{B}$, and $1 - \cos x = \frac{\sin^2 x}{1+\cos x}$.

e.g. $\sqrt{x+1} - \sqrt{x}$ for large x:

$$\sqrt{x+1} - \sqrt{x} = \frac{(\sqrt{x+1} - \sqrt{x})(\sqrt{x+1} + \sqrt{x})}{\sqrt{x+1} + \sqrt{x}} = \frac{(x+1) - x}{\sqrt{x+1} + \sqrt{x}} = \frac{1}{\sqrt{x+1} + \sqrt{x}}$$

Test at $x = 1000$ in 3-digit arithmetic ($\sqrt{1001} \to 31.6$, $\sqrt{1000} \to 31.6$):

- original: $31.6 - 31.6 = 0$ ✗ (every digit lost)
- rearranged: $\frac{1}{31.6 + 31.6} = \frac{1}{63.2} = 0.0158$ ✓ (true 0.015807)

▲ **Edge case: the sign of b decides which root is dangerous.** Lecture example $a = 1$, $b = -(10^8 + 10^{-8})$, $c = 1$ (roots 10^8 and 10^{-8}):

- $b < 0$ makes $-b > 0$, so the **minus** root $\frac{-b - \sqrt{b^2 - 4ac}}{2a}$ subtracts nearly-equal numbers
- naive formula in MATLAB: 1.49×10^{-8} ✗ (49% wrong)
- safe form $\frac{-2c}{b - \sqrt{b^2 - 4ac}}$: 10^{-8} ✓
- the plus root is fine here, because it adds two positive numbers

2.2 Taylor polynomial of degree n about x_0

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \cdots$$

Meaning: The polynomial that matches f and its first n derivatives at x_0. It's a good approximation **near** x_0.

e.g. $f(x) = \cos x$, degree 2, $x_0 = 0$:

- $f(x) = \cos x$, so $f(0) = 1$
- $f'(x) = -\sin x$, so $f'(0) = 0$
- $f''(x) = -\cos x$, so $f''(0) = -1$
- substitute: $P_2(x) = 1 + 0 \cdot x + \frac{-1}{2}x^2 = 1 - \frac{x^2}{2}$

▲ **Edge case: missing terms.** For $\cos x$ about 0, $f'''(0) = \sin 0 = 0$, so:

- $P_3(x) = 1 - \frac{x^2}{2} + 0 \cdot x^3 = P_2(x)$
- so you may use the R_3 remainder instead, which gives a sharper bound on $[0, \pi/3]$: $\frac{x^4}{24} \le \frac{(\pi/3)^4}{24} = 0.0501$ (vs 0.1658)

▲ **Edge case: f is already a polynomial of degree $\le n$.** Then $P_n = f$ exactly. For $f = x^2 + 3x$ about $x_0 = 1$ with $n = 2$:

- $f(1) = 4$, $f'(1) = 5$, $f''(1) = 2$
- $P_2 = 4 + 5(x-1) + (x-1)^2 = x^2 + 3x = f$ ✓

2.3 Taylor remainder (error)

$$R_n(x) = f(x) - P_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}, \qquad c \text{ between } x_0 \text{ and } x$$

Meaning: The exact error, but c is unknown. It uses the $(n+1)$th derivative.

e.g. Same $f = \cos x$, $n = 2$, $x_0 = 0$:

- the next derivative is $f'''(x) = \sin x$
- $R_2(x) = \frac{\sin c}{3!}(x - 0)^3 = \frac{\sin c}{6}x^3$, with c between 0 and x

▲ **Edge case: at $x = x_0$ the remainder is 0.** $R_n(x_0) = f^{(n+1)}(c) \cdot 0 = 0$, because $P_n(x_0) = f(x_0)$ by construction.

▲ **Edge case: c changes with x.** Every x has its own unknown c, so never try to solve for c; bound it instead.

2.4 Error bound recipe

$$|f(x) - P_n(x)| \le \frac{\max|f^{(n+1)}(c)|}{(n+1)!} \cdot \max|x - x_0|^{n+1}$$

Meaning: Replace each unknown by its worst case over the interval. The result is an upper bound on the error.

e.g. Same, for $x \in [0, \pi/3]$ (2025 Q2b):

- $c \in [0, \pi/3]$, where sin is increasing, so $|\sin c| \le \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$
- $|x|^3 \le \left(\frac{\pi}{3}\right)^3 = 1.1484$
- error $\le \frac{\sqrt{3}/2}{6} \times 1.1484 = 0.1658$

▲ **Edge case: the maximum may be inside the interval.** For $|\sin c|$ on $[0, \pi]$:

- endpoints: $\sin 0 = 0$ and $\sin \pi = 0$
- but the max is $\sin\frac{\pi}{2} = 1$ in the middle, so the bound must use 1

▲ **Edge case: interval on both sides of x_0.** For $x_0 = 0$ on $[-1, 2]$: $\max|x - x_0| = \max(1, 2) = 2$, not 1.

Remember

- Truncation error = from cutting maths short (finite terms, finite h). Roundoff error = from storing finitely many digits.
- Always state where c lies. Calculator in **radians**.

Week 3: ODEs & Euler (3 formulas)

3.1 Grid

$$h = \frac{b-a}{n}, \qquad t_i = a + ih, \qquad w_i \approx y(t_i)$$

Meaning: Chop $[a, b]$ into n steps of width h. The method outputs a **table** of values w_i, not a formula.

e.g. Solve on $[a, b] = [1, 2]$ with $n = 2$ steps:

- $h = \frac{2-1}{2} = 0.5$
- $t_0 = 1$, $t_1 = 1 + 0.5 = 1.5$, $t_2 = 1 + 2(0.5) = 2$
- the method produces $w_0, w_1, w_2 \approx y(1)$, $y(1.5)$, $y(2)$

▲ **Edge case: h must divide the interval evenly.** On $[0, 1]$:

- $h = 0.25$ gives $n = 4$ steps ✓
- $h = 0.3$ gives $n = 3.33$ ✗: you can't land exactly on $t = b$
- choose h (or n) so that $n = \frac{b-a}{h}$ is a whole number

3.2 Euler's method

$$w_0 = \alpha, \qquad w_{i+1} = w_i + h\, f(t_i, w_i)$$

Meaning: New value = old value + step × slope at the old point. You walk along the tangent line.

e.g. $y' = t + y^2$, $y(0) = 1$, $h = 0.1$:

- setup: $f(t, y) = t + y^2$, $t_0 = 0$, $w_0 = 1$
- slope at the start: $f(0, 1) = 0 + 1^2 = 1$
- step: $w_1 = w_0 + h\, f(t_0, w_0) = 1 + 0.1(1) = 1.1$
- so $y(0.1) \approx 1.1$

▲ **Edge case: zero starting slope.** $y' = -2ty$, $y(0) = 1$: $f(0, 1) = 0$, so $w_1 = w_0 = 1$. Euler doesn't move on the first step, even though the true curve does.

▲ **Edge case: h too large** (stability, good to know). $y' = -20y$, $y(0) = 1$ (exact $e^{-20t} \to 0$). Euler gives $w_{i+1} = (1 - 20h)\, w_i$:

- $h = 0.1$: factor -1, so $w = -1, 1, -1, 1, \ldots$ (oscillates forever)
- $h = 0.15$: factor -2, so $w = -2, 4, -8, 16, \ldots$ (blows up)
- fix: use a smaller h (here $h < 0.1$)

3.3 Local → global error

$$\text{local } O(h^{p+1}) \implies \text{global } O(h^p) \qquad \text{(Euler: local } O(h^2),\ \text{global } O(h))$$

Meaning: Local error = error made in **one** step, if you started that step exactly right; it's the dropped Taylor term. Global error = total error at the end. You take $\frac{b-a}{h}$ steps, so you lose one power of h.

e.g. $y' = y$, $y(0) = 1$ (exact $y = e^t$). Euler gives $w_1 = 1 + h$:

- one step, $h = 0.1$: $e^{0.1} - 1.1 = 0.0052$
- one step, $h = 0.05$: $e^{0.05} - 1.05 = 0.0013$ ($\div 4$, so the local error is $O(h^2)$)
- to $t = 1$, $h = 0.1$ (10 steps): $e - 1.1^{10} = 0.1245$
- to $t = 1$, $h = 0.05$ (20 steps): $e - 1.05^{20} = 0.0650$ ($\div 2$, so the global error is $O(h)$)

▲ **Edge case: zero local error.** If the true solution is a straight line, $y'' = 0$ and Euler is **exact**:

- $y' = 2$, $y(0) = 1$ (exact $y = 1 + 2t$)
- Euler: $w_{i+1} = w_i + 2h$, so $w_i = 1 + 2t_i = y(t_i)$ for any h

Remember

- "Order of a method" = order of the **global** error. Order p means halving h divides the error by 2^p.
- Derivation moves (all ODE methods): Taylor expand y about t_i → evaluate at t_{i+1} ($t_{i+1} - t_i = h$) → use $y' = f$ → drop $O(h^{\cdot})$ and replace y with w.
- Systems (SIR): apply Euler to every variable, using the **old** values on the right-hand side.

Week 4: Higher-order ODE methods (4 formulas)

4.1 Second-order Taylor method

$$w_{i+1} = w_i + h\, f(t_i, w_i) + \frac{h^2}{2} f'(t_i, w_i)$$

Meaning: Euler plus a curvature correction (the next Taylor term). Global $O(h^2)$.

e.g. $y' = y - t$, $y(0) = 2$, $h = 0.2$:

- $f(t, y) = y - t$, so $f(0, 2) = 2$
- $f'(t, y) = y' - 1 = (y - t) - 1$, so $f'(0, 2) = 1$
- $w_1 = 2 + 0.2(2) + \frac{0.2^2}{2}(1) = 2 + 0.4 + 0.02 = 2.42$
- exact $y(0.2) = 2.4214$

▲ **Edge case: exact for quadratic solutions.** $y' = 2t$, $y(0) = 0$ (exact $y = t^2$):

- $f = 2t$ and $f' = 2$
- $w_1 = 0 + h(0) + \frac{h^2}{2}(2) = h^2 = y(h)$ ✓
- exact because the dropped term involves $y''' = 0$

4.2 Total derivative (for Taylor 2)

$$f'(t, y) = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial y} \cdot f(t, y)$$

Meaning: y depends on t, so use the chain rule and replace y' with f. This is the most common lost mark and the most common MATLAB bug.

e.g. $f = y - t$:

- $\frac{\partial f}{\partial t} = -1$ and $\frac{\partial f}{\partial y} = 1$
- $f' = -1 + 1 \cdot f = -1 + (y - t) = y - t - 1$

Another: $f = 4t + y^2$ (this was a 2023 Q4 bug):

- $\frac{\partial f}{\partial t} = 4$ and $\frac{\partial f}{\partial y} = 2y$
- $f' = 4 + 2y \cdot f = 4 + 2y(4t + y^2)$, **not** $4 + 2y$

▲ **Edge case: f depends on only one variable.**

- only t: $f = t^2$ gives $f' = 2t$ (no chain-rule term, since $\frac{\partial f}{\partial y} = 0$)
- only y: $f = y^2$ gives $f' = 0 + 2y \cdot f = 2y \cdot y^2 = 2y^3$ (only the chain-rule term)

4.3 Modified Euler

$$k_1 = hf(t_i, w_i), \qquad k_2 = hf(t_i + h,\ w_i + k_1), \qquad w_{i+1} = w_i + \tfrac{1}{2}(k_1 + k_2)$$

Meaning: Do a trial Euler step, measure the slope at the predicted end, then average the start and end slopes. Global $O(h^2)$ with no f' needed.

e.g. $y' = t + y^2$, $y(0) = 1$, $h = 0.1$:

- $k_1 = 0.1\, f(0, 1) = 0.1(0 + 1) = 0.1$
- predicted end point: $(0 + 0.1,\ 1 + 0.1) = (0.1,\ 1.1)$
- $k_2 = 0.1\, f(0.1, 1.1) = 0.1(0.1 + 1.21) = 0.131$
- $w_1 = 1 + \dfrac{1}{2}(0.1 + 0.131) = 1.1155$

▲ **Edge case: f depends only on t.** Then Modified Euler is exactly the **trapezoidal rule** (Week 11): $w_{i+1} = w_i + \dfrac{h}{2}\big[f(t_i) + f(t_{i+1})\big]$. For $y' = 2t$, $y(0) = 0$, $h = 0.5$:

- $k_1 = 0.5(2 \times 0) = 0$
- $k_2 = 0.5(2 \times 0.5) = 0.5$
- $w_1 = 0 + \dfrac{1}{2}(0 + 0.5) = 0.25 = y(0.5)$ ✓ (the trapezoid is exact for linear f)

4.4 RK4

$$k_1 = hf(t_i, w_i), \qquad k_2 = hf\big(t_i + \tfrac{h}{2},\ w_i + \tfrac{k_1}{2}\big)$$
$$k_3 = hf\big(t_i + \tfrac{h}{2},\ w_i + \tfrac{k_2}{2}\big), \qquad k_4 = hf(t_i + h,\ w_i + k_3)$$
$$w_{i+1} = w_i + \tfrac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Meaning: Four slope samples (start, midpoint twice, end) weighted $1 : 2 : 2 : 1$. Global $O(h^4)$. Note that k_4 uses the **full** k_3.

e.g. $y' = 3t - \frac{y}{t}$, $y(1) = 2$, $h = 0.5$:

- $k_1 = 0.5\, f(1,\ 2) = 0.5(3 - 2) = 0.5$
- $k_2 = 0.5\, f(1.25,\ 2.25) = 0.5(3.75 - 1.8) = 0.975$
- $k_3 = 0.5\, f(1.25,\ 2.4875) = 0.5(3.75 - 1.99) = 0.88$
- $k_4 = 0.5\, f(1.5,\ 2.88) = 0.5(4.5 - 1.92) = 1.29$
- $w_1 = 2 + \dfrac{1}{6}\big(0.5 + 2(0.975) + 2(0.88) + 1.29\big) = 2 + \dfrac{5.5}{6} = 2.9167$ (exact 2.9167)

▲ **Edge case: f depends only on t.** Then $k_2 = k_3$ and RK4 is exactly **Simpson's rule**: $w_{i+1} = w_i + \dfrac{h}{6}\big[f(t_i) + 4f(t_i + \dfrac{h}{2}) + f(t_{i+1})\big]$. For $y' = 3t^2$, $y(0) = 0$, $h = 1$:

- $k_1 = 3(0)^2 = 0$
- $k_2 = k_3 = 3(0.5)^2 = 0.75$
- $k_4 = 3(1)^2 = 3$
- $w_1 = \dfrac{1}{6}(0 + 1.5 + 1.5 + 3) = 1 = y(1)$ ✓ (Simpson is exact for cubics)

▲ **Edge case (MATLAB):** k_4 uses the **full** k_3; writing `w+k3/2` was one of the bugs in 2025 Q4.

Remember

Method	Global order	Needs f'?
Euler	1	no
Taylor 2	2	yes
Modified Euler	2	no
RK4	4	no

- Modified Euler derivation: Taylor 2 + $f' \approx \frac{f(t_{i+1},y_{i+1})-f(t_i,y_i)}{h} \to w_{i+1} = w_i + \frac{h}{2}[f_i + f_{i+1}] \to$ replace w_{i+1} on the right-hand side with Euler's $w_i + hf_i$.

Week 5: Interpolation, Lagrange (2 formulas)

5.1 Lagrange form

$$P_n(x) = \sum_{i=0}^{n} L_{n,i}(x)\, y_i, \qquad L_{n,i}(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$$

Meaning: $L_{n,i}$ equals 1 at x_i and 0 at every other node. So the sum passes through every data point.

e.g. $f = \log x$, nodes $1, 2, 4$; estimate $\log 3$:

- y values: $\log 1 = 0$, $\log 2 = 0.6931$, $\log 4 = 1.3863$
- $L_{2,0}(3) = \dfrac{(3-2)(3-4)}{(1-2)(1-4)} = \dfrac{-1}{3}$
- $L_{2,1}(3) = \dfrac{(3-1)(3-4)}{(2-1)(2-4)} = \dfrac{-2}{-2} = 1$
- $L_{2,2}(3) = \dfrac{(3-1)(3-2)}{(4-1)(4-2)} = \dfrac{2}{6} = \dfrac{1}{3}$
- check: $-\dfrac{1}{3} + 1 + \dfrac{1}{3} = 1$ ✓
- $P_2(3) = -\dfrac{1}{3}(0) + 1(0.6931) + \dfrac{1}{3}(1.3863) = 1.1552$ (true 1.0986)

▲ **Edge case: repeated node.** If $x_i = x_j$, the denominator $(x_i - x_j) = 0$, so the nodes must be **distinct**.

▲ **Edge case: "degree at most n".** $(0,1), (1,3), (2,5)$ lie on a line, so $P_2(x) = 1 + 2x$ (the x^2 coefficient is 0). Three points don't always give a genuine quadratic.

▲ **Edge case: evaluating at a node** returns the data exactly: $P_2(x_1) = y_1$, because $L_{n,1}(x_1) = 1$ and the other basis polynomials are 0 there.

5.2 Interpolation remainder

$$f(x) - P_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)(x - x_1)\cdots(x - x_n), \qquad c \in [a, b]$$

Meaning: Like the Taylor remainder, but $(x - x_0)^{n+1}$ becomes a product over **all** nodes, so the error is zero at each node.

e.g. Same ($\log x$, nodes $1, 2, 4$, at $x = 3$):

- $f'''(x) = \dfrac{2}{x^3}$ is largest on $[1, 4]$ at $c = 1$, so $|f'''(c)| \le 2$
- product: $|(3-1)(3-2)(3-4)| = 2$
- bound: $|R_2(3)| \le \dfrac{2}{3!} \times 2 = 0.667$
- actual error: $|1.1552 - 1.0986| = 0.057 \le 0.667$ ✓

▲ **Edge case: extrapolation.** Same nodes $1, 2, 4$, but now at $x = 8$ (outside them):

- $P_2(8) = 0$, while $\log 8 = 2.079$ (error 2.08)
- product: $|(8-1)(8-2)(8-4)| = 168$, so the bound is $\dfrac{2}{6} \times 168 = 56$ (valid, but huge)
- outside the nodes, the product grows fast, and so does the error

▲ **Edge case: polynomial data.** If f is a polynomial of degree $\le n$, then $f^{(n+1)} = 0$ and the error is **exactly 0** (e.g. 3 points from x^2).

Remember

- **Uniqueness:** $n+1$ distinct points give exactly one polynomial of degree $\le n$. Lagrange, divided differences and forward differences all give the **same** polynomial.
- $n + 1$ points means degree n, which means the bound uses $f^{(n+1)}$.
- Check: $\sum_i L_{n,i}(x) = 1$.
- Relative error in code divides by $|f(a)|$ (the true value).

Week 6: Newton divided differences (2 formulas)

6.1 Divided difference

$$f[x_i] = y_i, \qquad f[x_i, \ldots, x_{i+k}] = \frac{f[x_{i+1}, \ldots, x_{i+k}] - f[x_i, \ldots, x_{i+k-1}]}{x_{i+k} - x_i}$$

Meaning: (Drop the first point) minus (drop the last point), divided by (last x minus first x). First differences are slopes; second differences are slopes of slopes.

e.g. Data $(-1, 3), (0, 1), (2, 3)$:

- $f[-1, 0] = \dfrac{1 - 3}{0 - (-1)} = -2$
- $f[0, 2] = \dfrac{3 - 1}{2 - 0} = 1$
- $f[-1, 0, 2] = \dfrac{f[0, 2] - f[-1, 0]}{2 - (-1)} = \dfrac{1 - (-2)}{3} = 1$

▲ **Edge case: order doesn't matter.** $f[2, 0] = \dfrac{y_0 - y_2}{0 - 2} = \dfrac{y_2 - y_0}{2 - 0} = f[0, 2]$. Divided differences are symmetric, but always divide by the last minus the first x **of the same list**.

▲ **Edge case: a repeated x** gives division by zero, so the nodes must be distinct.

6.2 Newton divided difference form

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \cdots$$

Meaning: The coefficients are the **top diagonal** of the table. Adding a point adds **one term**, and the old work is kept.

e.g. Same data. The top diagonal is $3,\ -2,\ 1$:

- $P_2(x) = 3 + (-2)(x + 1) + 1 \cdot (x + 1)(x - 0)$
- expand: $3 - 2x - 2 + x^2 + x = x^2 - x + 1$
- check: $P_2(2) = 4 - 2 + 1 = 3$ ✓

▲ **Edge case: the new point is already on the curve.** Add $(1, 1)$ to the data above (check: $P_2(1) = 1 - 1 + 1 = 1$):

- $f[2, 1] = \dfrac{1 - 3}{1 - 2} = 2$
- $f[0, 2, 1] = \dfrac{2 - 1}{1 - 0} = 1$
- $f[-1, 0, 2, 1] = \dfrac{1 - 1}{1 - (-1)} = 0$
- the new coefficient is **0**, so $P_3 = P_2$

Remember

- Nodes can be in any order; always divide by the outermost x's of the span.
- Exam: expand $f[x_0, x_1, x_2, x_3]$ by applying 6.1 repeatedly (one level per line).

Week 7: Newton forward differences (3 formulas)

7.1 Forward differences

$$\Delta y_i = y_{i+1} - y_i, \qquad \Delta^2 y_i = y_{i+2} - 2y_{i+1} + y_i, \qquad \Delta^3 y_i = y_{i+3} - 3y_{i+2} + 3y_{i+1} - y_i$$

Meaning: Plain subtractions down the table, with no division. Use them only for **equally spaced** data.

e.g. $y = 1, 2, 4, 8$ (from 2^x at $x = 0, 1, 2, 3$):

- Δy: $2 - 1 = 1$, $4 - 2 = 2$, $8 - 4 = 4$
- $\Delta^2 y$: $2 - 1 = 1$, $4 - 2 = 2$
- $\Delta^3 y$: $2 - 1 = 1$
- top diagonal: $y_0 = 1$, $\Delta y_0 = 1$, $\Delta^2 y_0 = 1$, $\Delta^3 y_0 = 1$

▲ **Edge case: polynomial data.** Degree-d data have constant Δ^d and zero Δ^{d+1}. For $y = x^2$ at $x = 0, 1, 2, 3$:

- y: 0, 1, 4, 9
- Δy: 1, 3, 5
- $\Delta^2 y$: 2, 2 (constant)
- $\Delta^3 y$: 0

▲ **Edge case: unequal spacing** (e.g. $x = 0, 1, 3$): forward differences don't apply, so use divided differences instead.

7.2 Newton forward difference form

$$P_n(x) = \sum_{k=0}^{n} \binom{s}{k} \Delta^k y_0 = y_0 + s\Delta y_0 + \frac{s(s-1)}{2!}\Delta^2 y_0 + \frac{s(s-1)(s-2)}{3!}\Delta^3 y_0 + \cdots, \qquad s = \frac{x - x_0}{h}$$

Meaning: The coefficients are the top diagonal of the difference table. **Convert $x \to s$ before evaluating.**

e.g. Same data, $h = 1$, so $s = \frac{x-0}{1} = x$:

- $P_3 = 1 + 1 \cdot s + \frac{s(s-1)}{2}(1) + \frac{s(s-1)(s-2)}{6}(1)$
- to estimate $2^{0.5}$: $x = 0.5$, so $s = 0.5$
- terms: $1 + 0.5 + \frac{(0.5)(-0.5)}{2} + \frac{(0.5)(-0.5)(-1.5)}{6} = 1 + 0.5 - 0.125 + 0.0625$
- $P_3 = 1.4375$ (true $\sqrt{2} = 1.4142$)

▲ **Edge case: s outside $[0, n]$ means extrapolation.** For the population data 1970–2010 ($s = 0, \ldots, 4$), 1950 is $s = -2$, which gives $P_4 = 3217$: more than in 1970, so unrealistic.

▲ **Edge case: x_0 must be the first node.** s is measured from x_0, and the top diagonal belongs to x_0.

7.3 Link to divided differences

$$f[x_0, x_1, \ldots, x_k] = \frac{\Delta^k y_0}{k!\, h^k}$$

Meaning: With equal spacing, divided differences become forward differences scaled by $k!\, h^k$. Proving it for $k = 2$ is an exam question.

e.g. $(1, 1), (2, 4), (3, 9)$ from $y = x^2$, with $h = 1$:

- $\Delta^2 y_0 = y_2 - 2y_1 + y_0 = 9 - 8 + 1 = 2$
- $f[x_0, x_1, x_2] = \frac{2}{2! \cdot 1^2} = 1$
- check: $f[1, 2] = 3$, $f[2, 3] = 5$, $f[1, 2, 3] = \frac{5-3}{3-1} = 1$ ✓ (the x^2 coefficient)

▲ **Edge case: don't forget h^k when $h \neq 1$.** For $(0, 0), (0.5, 0.25), (1, 1)$ from $y = x^2$, $h = 0.5$:

- $\Delta^2 y_0 = 1 - 2(0.25) + 0 = 0.5$
- $f[x_0, x_1, x_2] = \frac{0.5}{2! \times 0.5^2} = 1$ ✓
- forgetting h^2 gives $\frac{0.5}{2} = 0.25$ ✗

Remember

- Interpolation (inside the data) is reliable. **Extrapolation** (outside) is unreliable.
- $\binom{s}{k} = \frac{s(s-1)\cdots(s-k+1)}{k!}$ works for non-integer s.

Week 8: Bisection & fixed point (4 formulas)

8.1 Bracketing condition (from the IVT)

$$f \text{ continuous on } [a,b] \text{ and } f(a)\,f(b) < 0 \implies \text{a root in } [a,b]$$

Meaning: If the function changes sign, it must cross zero. Write: "opposite signs, so $[a,b]$ is a bracketing".

e.g. $f(x) = x^3 - 2x - 7$ on $[1,3]$ (2025 Q9a):

- $f(1) = 1 - 2 - 7 = -8 < 0$
- $f(3) = 27 - 6 - 7 = 14 > 0$
- opposite signs and f is continuous, so $[1,3]$ is a bracketing

▲ **Edge case: sign change but no root.** $f(x) = \dfrac{1}{x}$ on $[-1,1]$: $f(-1) = -1$ and $f(1) = 1$, but there's no root, because f jumps at $x = 0$. The IVT needs f to be **continuous**.

▲ **Edge case: no sign change, yet roots exist.** $f(x) = x^2 - 0.25$ on $[-1,1]$: $f(\pm 1) = 0.75$, both positive, but there are roots at ± 0.5 (two crossings cancel out).

8.2 Bisection

$$p = \frac{a+b}{2}; \quad \text{if } \operatorname{sign} f(p) = \operatorname{sign} f(a) \text{ then } a \leftarrow p, \text{ else } b \leftarrow p; \qquad |p_n - \text{root}| \le \frac{b-a}{2^n}$$

Meaning: Keep the half that still has the sign change. The approximation after n iterations is the nth midpoint. Guaranteed, but slow.

e.g. Same f, two iterations on $[1,3]$:

- iteration 1: $p = \dfrac{1+3}{2} = 2$, $f(2) = -3$ (same sign as $f(a)$), so $a \leftarrow 2$, giving $[2,3]$
- iteration 2: $p = \dfrac{2+3}{2} = 2.5$, $f(2.5) = 3.625$ (same sign as $f(b)$), so $b \leftarrow 2.5$, giving $[2,2.5]$
- estimate after 2 iterations: 2.5, with error $\le \dfrac{3-1}{2^2} = 0.5$

▲ **Edge case: $f(p) = 0$ exactly.** Stop: p is the root.

▲ **Edge case: how many iterations?** For error $\le 10^{-3}$ on $[1,3]$:

- you need $\dfrac{2}{2^n} \le 10^{-3}$, i.e. $2^n \ge 2000$
- so $n = 11$ iterations

▲ **Edge case: several roots in $[a,b]$.** Bisection finds **one** of them, and you can't choose which.

8.3 Fixed point iteration

$$x = g(x), \qquad x_{n+1} = g(x_n)$$

Meaning: Rewrite $f(x) = 0$ as $x = g(x)$ and feed each output back in. The answer is where $y = g(x)$ meets $y = x$.

e.g. Solve $x = \cos x$ with $g(x) = \cos x$, $x_0 = 0.5$:

- $x_1 = \cos 0.5 = 0.8776$
- $x_2 = \cos 0.8776 = 0.6390$
- $x_3 = \cos 0.6390 = 0.8027$
- …it zig-zags slowly in to 0.7391

▲ **Edge case: same equation, different g, opposite outcome.** For $0.9\cos x = \sqrt{x}$ (lecture), from $x_0 = 0.5$:

- $g_1(x) = \cos^{-1}\left(\dfrac{\sqrt{x}}{0.9}\right)$: 0.667, 0.444, 0.750, 0.276, 0.948, then undefined ✗
- $g_2(x) = 0.81\cos^2 x$: 0.624, 0.534, 0.600, … → 0.57 ✓

8.4 Fixed point theorem (convergence)

$$g(x) \in [a,b]\ \forall x \in [a,b] \quad \text{and} \quad |g'(x)| \le k < 1 \implies \text{unique fixed point, converges for any } x_0 \in [a,b]$$

Meaning: g must map the interval into itself and **shrink** distances. If $|g'| > 1$ near the root, try another rearrangement.

e.g. $g(x) = \cos x$ on $[0,1]$:

- maps into itself: $\cos 1 = 0.54 \le g(x) \le 1$, so $g(x) \in [0,1]$ ✓
- shrinks distances: $|g'(x)| = |\sin x| \le \sin 1 = 0.84 < 1$ ✓
- so the iteration converges for any $x_0 \in [0,1]$

▲ **Edge case: the theorem is sufficient, not necessary.** Take $g = \cos x$ on $[0,2]$:

- $g(2) = \cos 2 = -0.42 \notin [0,2]$, and $|g'(\frac{\pi}{2})| = 1$, which is not < 1
- so the theorem **doesn't apply**, yet the iteration still converges to 0.7391
- failing the test means "no guarantee", not "will diverge"

Week 9: Newton & secant (2 formulas + orders)

9.1 Newton's method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Meaning: Follow the tangent line at x_n down to the x-axis. Derivation: set $0 = f(x_n) + f'(x_n)(x_{n+1} - x_n)$.

e.g. $f(x) = x^3 - 2x - 7$, $x_0 = 2$ (2025 Q9c):

- $f(2) = 8 - 4 - 7 = -3$
- $f'(x) = 3x^2 - 2$, so $f'(2) = 10$
- $x_1 = 2 - \dfrac{-3}{10} = 2.3$

▲ **Edge case: $f'(x_n) = 0$.** For $f = x^2 - 2$ with $x_0 = 0$: $f'(0) = 0$, so you'd divide by zero (the tangent is horizontal and never meets the axis).

▲ **Edge case: cycling.** For $f = x^3 - 2x + 2$ with $x_0 = 0$:

- $x_1 = 0 - \dfrac{f(0)}{f'(0)} = 0 - \dfrac{2}{-2} = 1$
- $x_2 = 1 - \dfrac{f(1)}{f'(1)} = 1 - \dfrac{1}{1} = 0$
- back to the start, so it cycles $0, 1, 0, 1, \ldots$ forever; try another x_0

9.2 Secant method

$$x_{n+1} = x_n - f(x_n)\,\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$$

Meaning: Newton with f' replaced by the slope through the last two points. It needs 2 starting values and no derivative.

e.g. $f(x) = x^4 - 4x^3 + 2x + 3$, $x_0 = 1$, $x_1 = 2$ (2024 Q9c):

- $f(1) = 1 - 4 + 2 + 3 = 2$
- $f(2) = 16 - 32 + 4 + 3 = -9$
- $x_2 = 2 - (-9) \cdot \dfrac{2-1}{-9-2} = 2 - \dfrac{9}{11} = \dfrac{13}{11} = 1.1818$

▲ **Edge case: $f(x_n) = f(x_{n-1})$.** For $f = x^2 - 2$ with $x_0 = -1$, $x_1 = 1$: $f(-1) = f(1) = -1$, so the secant line is horizontal and you'd divide by zero.

▲ **Edge case: no bracket needed, so no guarantee.** Unlike bisection, the secant iterates can leave the starting interval.

9.3 Convergence orders

$$|x_{n+1}-p| \approx \lambda |x_n - p|^r : \quad r=1 \text{ (bisection, fixed point)}, \quad r=\tfrac{1+\sqrt{5}}{2} \approx 1.618 \text{ (secant)}, \quad r=2 \text{ (Newton)}$$

Meaning: $r=2$ means the number of correct digits roughly **doubles** each step. Secant is slower because it approximates the derivative.

e.g. Newton on $x-\cos x=0$ from $x_0=0.5$ (root 0.739085):

- $x_1=0.755222$: error 1.6×10^{-2}
- $x_2=0.739142$: error 5.7×10^{-5}
- $x_3=0.739085$: error 7×10^{-10}
- the error is roughly squared each step, so the correct digits double

▲ **Edge case: a repeated root slows Newton down.** For $f=x^2$ (double root at 0) from $x_0=1$:

- $x_{n+1}=x_n-\dfrac{x_n^2}{2x_n}=\dfrac{x_n}{2}$, giving 0.5, 0.25, 0.125, ...
- the error only **halves** each step, so convergence is linear, not quadratic

Remember

- Deriving an iteration for $\sqrt[k]{a}$: choose $f(x)=x^k-a$, apply Newton or secant, simplify.
- For secant, use the identities $p^2-q^2=(p-q)(p+q)$ and $p^3-q^3=(p-q)(p^2+pq+q^2)$.
- Heron: $x_{n+1}=\frac{1}{2}\left(x_n+\frac{a}{x_n}\right)$ is Newton on x^2-a.
- In code: `while` uses `> tol` (keep going), and `if … break` uses `<= tol` (stop).

Week 10: Numerical differentiation (5 formulas)

10.1 Forward difference (1st order)

$$f'(x_0)=\frac{f(x_0+h)-f(x_0)}{h}-\frac{h}{2}f''(c)$$

Meaning: Slope to the point on the right. Use it at the **left end** of a table.

e.g. Table x: 0.15, 0.3, 0.45; f: 1.3516, 1.3043, 1.2419 (2025 Q12a):

- $x_0=0.15$ is the left end, so use forward with $h=0.15$
- $f'(0.15)\approx\dfrac{f(0.3)-f(0.15)}{0.15}=\dfrac{1.3043-1.3516}{0.15}=\dfrac{-0.0473}{0.15}=-0.3153$

▲ **Edge case: exact for straight lines.** For $f=3x+1$: $\dfrac{f(x_0+h)-f(x_0)}{h}=\dfrac{3h}{h}=3$ for **any** h (the error term has $f''=0$).

▲ **Edge case: h too small.** Below about $h\approx10^{-8}$, roundoff (catastrophic cancellation) takes over, so a smaller h is **not** always better.

10.2 Backward difference (1st order)

$$f'(x_0) = \frac{f(x_0) - f(x_0 - h)}{h} + \frac{h}{2} f''(c)$$

Meaning: Slope to the point on the left. Use it at the **right end** of a table.

e.g. Table x: 1, 1.1, 1.2; f: 0.5679, 0.6542, 0.7286 (2023 Q12a):

- $x_0 = 1.2$ is the right end, so use backward with $h = 0.1$
- $f'(1.2) \approx \dfrac{0.7286 - 0.6542}{0.1} = \dfrac{0.0744}{0.1} = 0.7440$

▲ **Edge case: opposite error signs.** The backward error $(+\frac{h}{2} f'')$ and the forward error $(-\frac{h}{2} f'')$ have opposite signs, so their **average is exactly the central difference**. For $\sin x$ at $x_0 = 1$, $h = 0.1$:

- forward 0.4970, backward 0.5820
- average 0.5395 = central difference (true 0.5403)

10.3 Central difference for f' (2nd order)

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0 - h)}{2h} - \frac{h^2}{6} f'''(c)$$

Meaning: Slope across both neighbours. More accurate, but needs a point on each side.

e.g. Same table; find $f'(1.1)$ (2023 Q12b):

- $x_0 = 1.1$ has neighbours on both sides, so use central with $h = 0.1$
- $f'(1.1) \approx \dfrac{f(1.2) - f(1.0)}{2(0.1)} = \dfrac{0.7286 - 0.5679}{0.2} = 0.8035$

▲ **Edge case: exact for quadratics** (the error term has $f''' = 0$). For $f = x^2$, $x_0 = 1$, $h = 0.5$:

- central: $\dfrac{1.5^2 - 0.5^2}{2(0.5)} = \dfrac{2}{1} = 2 = f'(1)$ ✓
- forward: $\dfrac{1.5^2 - 1}{0.5} = 2.5$ ✗

▲ **Edge case: unequal spacing.** If the points on either side are at different distances from x_0, this formula doesn't apply.

10.4 Central difference for f'' (2nd order)

$$f''(x_0) = \frac{f(x_0 - h) - 2f(x_0) + f(x_0 + h)}{h^2} - \frac{h^2}{12} f^{(4)}(c)$$

Meaning: The "slope of slopes". Derivation: **add** the degree-3 Taylor expansions at $x_0 \pm h$, then merge the two c's with the IVT.

e.g. First table (x: 0.15, 0.3, 0.45); find $f''(0.3)$ (2025 Q12b):

- $h = 0.15$, so $h^2 = 0.0225$
- numerator: $1.3516 - 2(1.3043) + 1.2419 = -0.0151$
- $f''(0.3) \approx \dfrac{-0.0151}{0.0225} = -0.6711$

▲ **Edge case: exact for cubics** (the error term has $f^{(4)} = 0$). For $f = x^3$, $x_0 = 1$, $h = 0.5$:

- $\dfrac{0.5^3 - 2(1) + 1.5^3}{0.5^2} = \dfrac{0.125 - 2 + 3.375}{0.25} = 6 = f''(1)$ ✓

10.5 Second-order forward difference

$$f'(x_0) \approx \frac{-3f(x_0) + 4f(x_0 + h) - f(x_0 + 2h)}{2h}$$

Meaning: Second-order accuracy using only points to the right. Derivation: $4f(x_0 + h) - f(x_0 + 2h)$ cancels the h^2 terms.

e.g. $f(x) = \log x$, $x_0 = 2$, $h = 0.1$ (exact $f'(2) = 0.5$):

- values: $f(2) = 0.693147$, $f(2.1) = 0.741937$, $f(2.2) = 0.788457$
- second-order forward: $\dfrac{-3(0.693147) + 4(0.741937) - 0.788457}{0.2} = 0.49925$
- plain forward: $\dfrac{0.741937 - 0.693147}{0.1} = 0.4879$ (an error about 16× bigger)

▲ **Edge case: exact for quadratics.** For $f = x^2$, $x_0 = 1$, $h = 0.5$:

- $\dfrac{-3(1) + 4(1.5^2) - 2^2}{2(0.5)} = \dfrac{-3 + 9 - 4}{1} = 2 = f'(1)$ ✓

Remember

- Derivations: **subtract** the expansions to get f' (even terms cancel); **add** them to get f'' (odd terms cancel).
- Choosing h: too large gives **truncation** error; too small gives **roundoff** error (catastrophic cancellation).

Week 11: Numerical integration (4 formulas)

11.1 Trapezoidal rule (one strip)

$$\int_{x_0}^{x_1} f\,dx \approx \frac{h}{2}\big[f(x_0) + f(x_1)\big]$$

Meaning: The area of the trapezoid under the straight line (the degree-1 interpolant).

e.g. $\int_1^2 \frac{1}{x}\,dx$ with one strip ($h = 1$):

- $f(1) = 1$ and $f(2) = 0.5$
- $\frac{1}{2}[1 + 0.5] = 0.75$
- exact $\log 2 = 0.6931$: one straight line is crude

▲ **Edge case: convex vs concave.** For one strip, the error is $-\frac{h^3}{12}f''(c)$:

- convex ($f'' > 0$), e.g. $\frac{1}{x}$: the trapezoid **over**-estimates ($0.75 > 0.6931$)
- concave ($f'' < 0$), e.g. $\sin x$ on $[0, \pi]$: it **under**-estimates ($1.8961 < 2$)
- straight line ($f'' = 0$): **exact**

11.2 Composite trapezoidal rule

$$\int_a^b f\,dx = \frac{h}{2}\Big[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\Big] - \frac{(b-a)h^2}{12} f''(c)$$

Meaning: Weights $1, 2, 2, \ldots, 2, 1$. Second order, and exact for polynomials of degree ≤ 1.

e.g. Same integral, $n = 4$, $h = 0.25$:

- x: 1, 1.25, 1.5, 1.75, 2
- f: 1, 0.8, 0.6667, 0.5714, 0.5
- bracket: $1 + 2(0.8 + 0.6667 + 0.5714) + 0.5 = 5.5762$
- $\dfrac{0.25}{2} \times 5.5762 = 0.6970$ (error 0.004)

▲ **Edge case: n counts strips, not points.** 5 points means $n = 4$ and $h = \dfrac{b-a}{4}$; using $n = 5$ is a classic slip.

▲ **Edge case: unequal spacing.** The formula needs a single h; otherwise add up the individual trapezoids yourself.

11.3 Simpson's rule (one pair of strips)

$$\int_{x_0}^{x_2} f\,dx \approx \frac{h}{3}\big[f(x_0) + 4f(x_1) + f(x_2)\big]$$

Meaning: The area under the parabola (the degree-2 interpolant) through 3 points.

e.g. Same integral, one pair of strips ($h = 0.5$):

- $f(1) = 1$, $f(1.5) = \dfrac{2}{3}$, $f(2) = 0.5$
- bracket: $1 + 4\left(\dfrac{2}{3}\right) + 0.5 = 4.1667$
- $\dfrac{0.5}{3} \times 4.1667 = 0.6944$

▲ **Edge case: exact for cubics too, not just quadratics.** $\int_0^2 x^3\,dx = 4$, and with $h = 1$:

- $\dfrac{1}{3}\big[0 + 4(1) + 8\big] = 4$ ✓ (the error term has $f^{(4)} = 0$)

11.4 Composite Simpson's rule (n even)

$$\int_a^b f\,dx = \frac{h}{3}\Big[f(x_0) + 4\sum_{\text{odd } i} f(x_i) + 2\sum_{\text{even interior } i} f(x_i) + f(x_n)\Big] - \frac{(b-a)h^4}{180} f^{(4)}(c)$$

Meaning: Weights $1, 4, 2, 4, \ldots, 4, 1$. Fourth order, and exact for polynomials of degree ≤ 3.

e.g. Same integral, $n = 4$, $h = 0.25$:

- weights $1, 4, 2, 4, 1$ on $f = 1, 0.8, 0.6667, 0.5714, 0.5$
- bracket: $1 + 4(0.8 + 0.5714) + 2(0.6667) + 0.5 = 8.3190$
- $\dfrac{0.25}{3} \times 8.3190 = 0.6933$ (error 0.0001, against 0.004 for the trapezoid)

▲ **Edge case: n must be even.**

- 4 points means $n = 3$ (odd), so Simpson can't be used; use the trapezoidal rule instead
- 6 points means $n = 5$: same problem

▲ **Edge case (MATLAB):** `x(i)` is x_{i-1}, so the **even** MATLAB indices `x(2)`, `x(4)`, … are the odd abscissas and get weight **4**.

Remember

- Derivations: integrate the Newton forward difference interpolant with $x = x_0 + sh$, $dx = h\,ds$, s from 0 to 1 (trapezoid) or 0 to 2 (Simpson).
- In MATLAB, n subintervals means $n + 1$ points: `x(1)` is x_0 and `x(n+1)` is x_n.

Week 12: Linear systems & LU (4 formulas)

12.1 Forward substitution (lower triangular)

$$x_i = \frac{b_i - \sum_{j<i} a_{ij} x_j}{a_{ii}}, \qquad i = 1, 2, \ldots, n$$

Meaning: Row 1 gives x_1 directly. Substitute it into row 2 to get x_2, and so on **downwards**.

e.g. Solve (2025 Q15a):

$$\begin{bmatrix} 3 & 0 \\ -2 & 5 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 12 \\ -23 \end{bmatrix}$$

- row 1: $3x_1 = 12$, so $x_1 = 4$
- row 2: $-2(4) + 5x_2 = -23$, so $5x_2 = -15$ and $x_2 = -3$

▲ **Edge case: zero on the diagonal.** In the system below, row 1 says $0 \cdot x_1 = 1$, which is impossible:

$$\begin{bmatrix} 0 & 0 \\ 1 & 2 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

- a triangular system has a unique solution only if **every diagonal entry is nonzero**

12.2 Backward substitution (upper triangular)

$$x_i = \frac{b_i - \sum_{j>i} a_{ij} x_j}{a_{ii}}, \qquad i = n, \ldots, 1$$

Meaning: The last row gives x_n directly. Work **upwards**.

e.g. Solve:

$$\begin{bmatrix} 2 & 1 \\ 0 & 4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

- last row: $4x_2 = 8$, so $x_2 = 2$
- row 1: $2x_1 + 1(2) = 5$, so $x_1 = 1.5$

▲ **Edge case: zero diagonal entry, infinitely many solutions.**

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

- row 2 says $0 = 0$ (no information), so x_2 is free
- row 1 gives $x_1 = 2 - x_2$, which works for every choice of x_2

12.3 Solving with LU

$$A = LU: \qquad L\mathbf{z} = \mathbf{b} \text{ (forward)}, \quad \text{then} \quad U\mathbf{x} = \mathbf{z} \text{ (backward)}$$

Meaning: Factorise once (the expensive part), then each new $\mathbf{b}$ costs only two cheap triangular solves.

e.g. 2024 Q15b: $\mathbf{z} = [2, -8, -4]^T$ is given, so solve $U\mathbf{x} = \mathbf{z}$ with

$$U = \begin{bmatrix} 2 & 1 & -4 \\ 0 & -3 & 2 \\ 0 & 0 & 4 \end{bmatrix}$$

- row 3: $4x_3 = -4$, so $x_3 = -1$
- row 2: $-3x_2 + 2(-1) = -8$, so $x_2 = 2$
- row 1: $2x_1 + 2 - 4(-1) = 2$, so $x_1 = -2$
- $\mathbf{x} = [-2,\ 2,\ -1]^T$

▲ **Edge case: a new right-hand side.** Reuse the **same** L and U, and redo only the two substitutions. Never refactorise for a new $\mathbf{b}$.

▲ **Edge case: order matters.** Solve $L\mathbf{z} = \mathbf{b}$ **first** (forward), then $U\mathbf{x} = \mathbf{z}$ (backward). $\mathbf{z}$ is not the answer.

12.4 Doolittle 3×3 (L has 1s on the diagonal)

$$u_{1j} = a_{1j}, \quad \ell_{i1} = \frac{a_{i1}}{u_{11}}, \quad u_{22} = a_{22} - \ell_{21}u_{12}, \quad u_{23} = a_{23} - \ell_{21}u_{13}$$

$$\ell_{32} = \frac{a_{32} - \ell_{31}u_{12}}{u_{22}}, \qquad u_{33} = a_{33} - \ell_{31}u_{13} - \ell_{32}u_{23}$$

Meaning: All of these come from one rule: $a_{ij} = (\text{row } i \text{ of } L) \cdot (\text{column } j \text{ of } U)$. Order: row of U, column of L, row of U, ...

e.g. 2024 Q15a, with $\ell_{21} = -2$ and $\ell_{31} = 3$ given:

$$A = \begin{bmatrix} 2 & 1 & -4 \\ -4 & -5 & 10 \\ 6 & 6 & -10 \end{bmatrix}, \qquad u_{23} = 2$$

- $u_{22} = a_{22} - \ell_{21}u_{12} = -5 - (-2)(1) = -3$
- $\ell_{32} = \dfrac{a_{32} - \ell_{31}u_{12}}{u_{22}} = \dfrac{6 - 3(1)}{-3} = -1$
- $u_{33} = a_{33} - \ell_{31}u_{13} - \ell_{32}u_{23} = -10 - 3(-4) - (-1)(2) = 4$

▲ **Edge case: zero pivot.** The matrix below is invertible, but $u_{11} = 0$, so $\ell_{21} = \dfrac{1}{0}$ and LU (without pivoting) fails:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$$

- swapping the two rows (**pivoting**) fixes it

Remember

- A zero or tiny pivot u_{jj} makes LU unstable, which **pivoting** (row swaps) fixes (not covered in detail).

Week 13: SPD & Cholesky (3 formulas)

13.1 Symmetric positive definite

$$A^T = A \qquad \text{and} \qquad \mathbf{x}^T A\mathbf{x} > 0 \ \text{ for all } \mathbf{x} \neq \mathbf{0}$$

Meaning: Show it by expanding $\mathbf{x}^T A\mathbf{x}$ and completing the square into a sum of squares. For a 2×2 matrix: $\mathbf{x}^T A\mathbf{x} = ax_1^2 + 2bx_1x_2 + cx_2^2$.

e.g. $A = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$ (2025 Q16a):

- $A\mathbf{x} = \begin{bmatrix} x_1 - x_2 \\ -x_1 + 2x_2 \end{bmatrix}$
- $\mathbf{x}^T A\mathbf{x} = x_1(x_1 - x_2) + x_2(-x_1 + 2x_2) = x_1^2 - 2x_1x_2 + 2x_2^2$
- complete the square: $= (x_1 - x_2)^2 + x_2^2$
- this is > 0 unless $x_1 = x_2 = 0$, so A is positive definite

▲ **Edge case: positive entries don't make a matrix PD.** For $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $\mathbf{x} = (1, -1)$:

- $\mathbf{x}^T A\mathbf{x} = 1 - 4 + 1 = -2 < 0$, so A is symmetric but **not** PD

▲ **Edge case: ≥ 0 is not enough.** $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ gives $\mathbf{x}^T A\mathbf{x} = (x_1 + x_2)^2$, which is 0 for $\mathbf{x} = (1, -1) \neq \mathbf{0}$. That's positive **semi**-definite, not PD.

13.2 Cholesky 3×3 ($A = LL^T$)

$$\ell_{11} = \sqrt{a_{11}}, \quad \ell_{21} = \frac{a_{21}}{\ell_{11}}, \quad \ell_{31} = \frac{a_{31}}{\ell_{11}}, \quad \ell_{22} = \sqrt{a_{22} - \ell_{21}^2}$$

$$\ell_{32} = \frac{a_{32} - \ell_{31}\ell_{21}}{\ell_{22}}, \qquad \ell_{33} = \sqrt{a_{33} - \ell_{31}^2 - \ell_{32}^2}$$

Meaning: Diagonal entry: $\sqrt{\text{entry} - \text{sum of squares so far}}$. Below the diagonal: (entry − dot product so far) ÷ diagonal. Work column by column.

e.g. Factorise:

$$A = \begin{bmatrix} 16 & 4 & 8 \\ 4 & 5 & -4 \\ 8 & -4 & 22 \end{bmatrix}$$

- column 1: $\ell_{11} = \sqrt{16} = 4$, $\ell_{21} = \frac{4}{4} = 1$, $\ell_{31} = \frac{8}{4} = 2$
- column 2: $\ell_{22} = \sqrt{5 - 1^2} = 2$, $\ell_{32} = \frac{-4 - (2)(1)}{2} = -3$
- column 3: $\ell_{33} = \sqrt{22 - 2^2 - (-3)^2} = \sqrt{9} = 3$

▲ **Edge case: Cholesky fails.** For $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$:

- $\ell_{11} = 1$ and $\ell_{21} = 2$
- $\ell_{22} = \sqrt{1 - 4} = \sqrt{-3}$ ✗, so A is not PD; switch to LU

▲ **Edge case: zero under the root.** For $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$: $\ell_{22} = \sqrt{1 - 1} = 0$, and any later entry would divide by it, so A is not PD (it's singular).

13.3 Solving with Cholesky

$$L\mathbf{z} = \mathbf{b} \text{ (forward)}, \quad \text{then} \quad L^T\mathbf{x} = \mathbf{z} \text{ (backward)}$$

Meaning: Same idea as LU, with $U = L^T$.

e.g. Same A and L, with $\mathbf{b} = [20, 17, 28]^T$.

Forward, $L\mathbf{z} = \mathbf{b}$:

- $4z_1 = 20$, so $z_1 = 5$
- $z_1 + 2z_2 = 17$, so $z_2 = 6$
- $2z_1 - 3z_2 + 3z_3 = 28$, so $z_3 = 12$

Backward, $L^T\mathbf{x} = \mathbf{z}$:

- $3x_3 = 12$, so $x_3 = 4$
- $2x_2 - 3x_3 = 6$, so $x_2 = 9$
- $4x_1 + x_2 + 2x_3 = 5$, so $x_1 = -3$

▲ **Edge case: the second solve goes bottom-up.** L^T is **upper** triangular, so solve $L^T\mathbf{x} = \mathbf{z}$ from the last row up (start with x_3). Working top-down is a common slip.

Remember

- Cholesky needs A **symmetric and positive definite**. It does about half the work of LU, needs no pivoting, and is always stable.
- Cholesky **succeeds** (every square root has a positive argument) ⇔ A is SPD. A negative under the root means A is not SPD, so switch to LU.

Exam-day extras

"Which is more accurate?"

Name the orders: "(b) is second order, $O(h^2)$, whereas (a) is first order, $O(h)$, so (b) has the smaller error for the same h."

MATLAB bug checklist (3 questions, 18 marks)

Variables initialised? · `h=(b-a)/n`? · arguments in order `f(t,w)`? · `.* ./ .^` on vectors? · right coefficients ($k_3/2$, `2*`, $h/2$ vs $h/3$, `/6`)? · indices `x(1)`…`x(n+1)`? · stop conditions (`> tol` vs `<= tol`)? · Newton `f/fdash` not upside down? · relative error divides by `abs(true)`?